SDAR Study Tour

Learning map – Self-Distilled Agentic Reinforcement Learning (Lu et al. 2026, arXiv:2605.15155)

1. Reader's contract

This learning map is for someone who wants to understand SDAR at the level needed to reproduce the math, critique the design, and propose extensions. Three skill-level entry points: **Beginner** (6–8 hours, read all of §2), **Intermediate** (3–4 hours, focus on §3), **Advanced** (1–2 hours, jump to §3 and §4). The full chain prerequisite is Chain Ch 25, 27, 28, 30, 31. Companion artifacts: 14 paper concept notes, 4 improvement concept notes, 18 runnable Python demos, 2 LaTeX proofs.

2. Foundations walk

These five chapters from the foundations chain are prerequisite. Read in order.

Chain Ch 25 – Causal LM as MDP. The per-token autoregressive decoding loop is a Markov decision process: state = prompt + tokens-so-far, action = next token. Why this paper needs it: every per-token decision in SDAR is one transition; the multi-turn agentic setting wraps this MDP in an outer environment loop. *Pacing: 30 min, conceptual.*

Chain Ch 27 – Tiny GPT pre-training. The decoder-only Transformer architecture. Why this paper needs it: the Qwen 2.5 backbone follows this recipe; you need to be fluent enough to read the SDAR loss equations as gradients on the parameters of such a model. *Pacing: 60 min, mostly review for advanced readers.*

Chain Ch 28 – SFT, RLHF, DPO. Supervised fine-tuning, KL-regularized RL from human feedback, direct preference optimization. Why this paper needs it: SDAR's distillation loss is structurally a KL-to-reference, the same primitive as RLHF's KL-to-SFT regularizer. The asymmetric-trust observation parallels DPO's conservatism on negatives. *Pacing: 90 min, central.*

Chain Ch 30 – Maximum-entropy RL. Soft Q-learning, the entropy bonus, the Lagrangian dual of constrained MDPs. Why this paper needs it: the gated SDAR loss IS a Lagrangian-style auxiliary, with λ_{SDAR} as the multiplier on a per-token trust constraint. Improvement 104 makes this formal. *Pacing: 60 min for the dual-form derivation alone.*

Chain Ch 31 – Policy gradients, GRPO, RLHF-DPO bridge. REINFORCE, PPO, GRPO, the policy-gradient theorem, single-sample estimators. Why this paper needs it: SDAR's RL backbone is GRPO from this chapter verbatim (Equation 2 of the paper). The single-sample teacher-student gap estimator uses the same expectation-replacement trick as the proof of the policy-gradient theorem. *Pacing: 90–120 min, core.*

3. Paper concepts walk

A topological order for the 14 concept notes:

1. Multi-turn agentic MDP – set up the outer loop.
2. Trajectory-level RL reward – sparse supervision is the pain.
3. GRPO baseline – RL backbone.
4. OPSD – on-policy self-distillation primitive.
5. Teacher branch with privileged context – what makes "self" non-trivial.
6. Per-token reverse KL – mode-seeking, $D_{\mathrm{KL}}(\pi_\theta \| \pi_T)$.
7. Single-sample gap estimator – $\Delta_t = \log \pi_T(y_t \mid s_t^+) - \log \pi_\theta(y_t \mid s_t)$.
8. Multi-turn instability observation (Obs 1) – compounding covariate shift.
9. Asymmetric trust observation (Obs 2) – positive vs negative gaps.
10. Sigmoid token-level gate – $g_t = \mathrm{sg}(\sigma(\beta \cdot \mathrm{score}_t))$.
11. Entropy gating – $g_t = \sigma(\beta h_t)$.
12. Gap gating – $g_t = \sigma(\beta \Delta_t)$ (centerpiece).
13. Soft-OR gating – $g_t = \sigma(\beta[1 - (1 - h_t)(1 - \Delta_t)])$.
14. SDAR combined objective – $\mathcal{L}(\theta) = \mathcal{L}_{\mathrm{GRPO}}(\theta) + \lambda_{\mathrm{SDAR}} \cdot \mathcal{L}_{\mathrm{SDAR}}(\theta)$.

4. Improvements walk

101 – Tight bias bound for gap gating. The paper's §7 sketch argues "lower bias" without a closed form. We state and prove a quantitative bound: for sub-Gaussian Δ_t with parameter σ_Δ,

$$|B(s_t)| \leq \tfrac{1}{2}\,\mathbb{E}[|\Delta_t|] + O(\beta \sigma_\Delta^2).$$

Validation mode: PROOF – see `proofs/gating-bias-bound.tex`.

102 – Adaptive gating threshold via running quantile. Replace fixed β with a per-batch standardization of Δ_t using a running median and IQR. The hyperparameter shifts from "absolute gate sharpness" to "fraction of tokens to trust". **Validation mode: MEASUREMENT** – see `improvements/adaptive-gate.py`.

103 – Sensitivity sweep of λ_{SDAR}. How flat is the loss vs λ landscape? Does an exponential anneal $\lambda(t) = \lambda_0 e^{-t/\tau}$ outperform a constant? Envelope theorem makes per-λ sensitivity identifiable from a single trained checkpoint. **Validation mode: MEASUREMENT** – see `improvements/lambda-sweep.py`.

104 – SDAR as a per-token trust-constraint Lagrangian. The SDAR combined loss is the Lagrangian of "maximize GRPO objective subject to $\mathbb{E}_t[g_t \Delta_t] \leq \delta$". λ_{SDAR} is the (fixed) Lagrange multiplier. The connection imports primal-dual convergence theorems from the constrained-MDP literature. **Validation mode: PROOF** – see `proofs/sdar-as-constrained-rl.tex`.

5. What to do next

1. **(15 min)** Run all 14 paper concept Python scripts. The printed outputs together form a guided narrative of the paper's mechanics.
2. **(2 hr)** Open `tour.ipynb`, execute every cell, read the proofs. Goal: write the SDAR loss from memory, derive its gradient, explain the sigmoid choice unprompted.
3. **(1 day)** Implement improvement 102 (adaptive gating) and run a short ablation. A negative result is also publishable.